PROJECT REPORT

Secure Office Network Using VLANs and Inter-VLAN Routing

Cisco Packet Tracer | Router-on-a-Stick | Intermediate | Individual project

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1. Executive Summary

This report documents the design, build and testing of a segmented office network in Cisco Packet Tracer. Three departments, Human Resources (HR), Information Technology (IT) and Finance, each have their own VLAN and subnet. Every department is spread across two switches, so the network depends on 802.1Q trunking. A single router connects the VLANs using subinterfaces on one physical port (Router-on-a-Stick), and a shared server on its own routed subnet extends the design.

All connectivity tests passed. The failure test showed that same-VLAN traffic keeps working without the router while inter-VLAN traffic stops, and a packet-level trace in Simulation Mode confirmed how MAC addresses, VLAN tags and IP addresses change at each hop. Section 3 is written as a step-by-step guide, so the network can be rebuilt from scratch.

Table 1: Results at a glance

Test	Outcome	Detail in
A. Same-VLAN	Passed: all three department pairs reachable, 4 of 4 replies, TTL 128	Section 5.1
B. Inter-VLAN	Passed: HR to IT, HR to Finance and Finance to HR reachable, TTL 127 (one router hop)	Section 5.2
C. Router failure	Passed: same-VLAN unaffected, inter-VLAN fully blocked, restored after no shutdown	Section 5.3
D. Packet-flow trace	Completed: ARP, 802.1Q tagging, MAC rewrite and unchanged IPs all observed	Section 5.4
E. Server reachability	Passed: every VLAN reaches 192.168.40.10, TTL 127	Section 5.5

2. Network Design

2.1 Topology

Router0 connects to Switch1 through one trunk link, and Switch1 connects to Switch2 through a second trunk. Each switch carries one PC from every department, which is why the inter-switch trunk is essential. The server connects directly to the router on its own port.

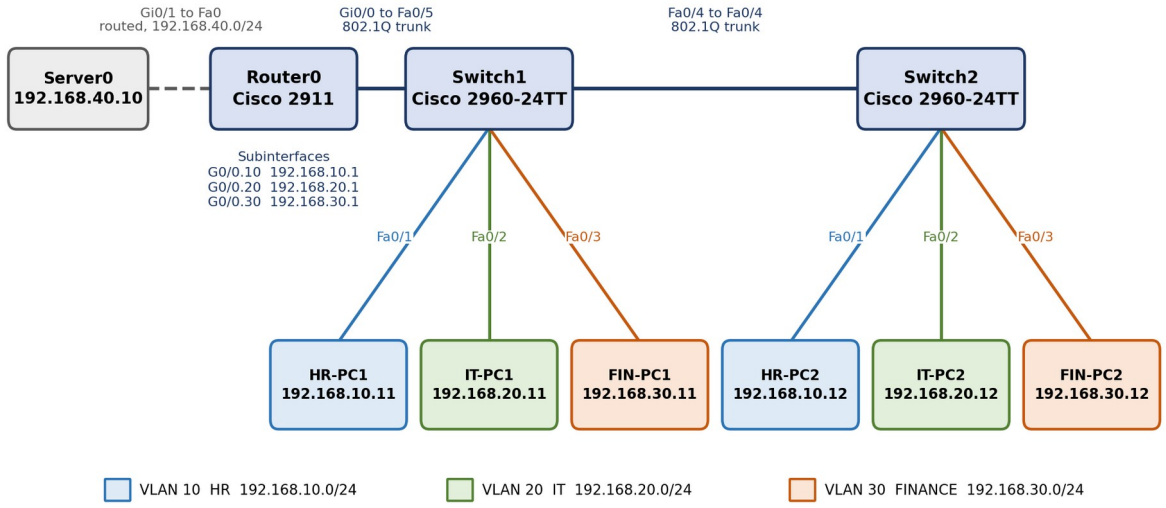


Figure 1: Logical topology with port connections and addressing

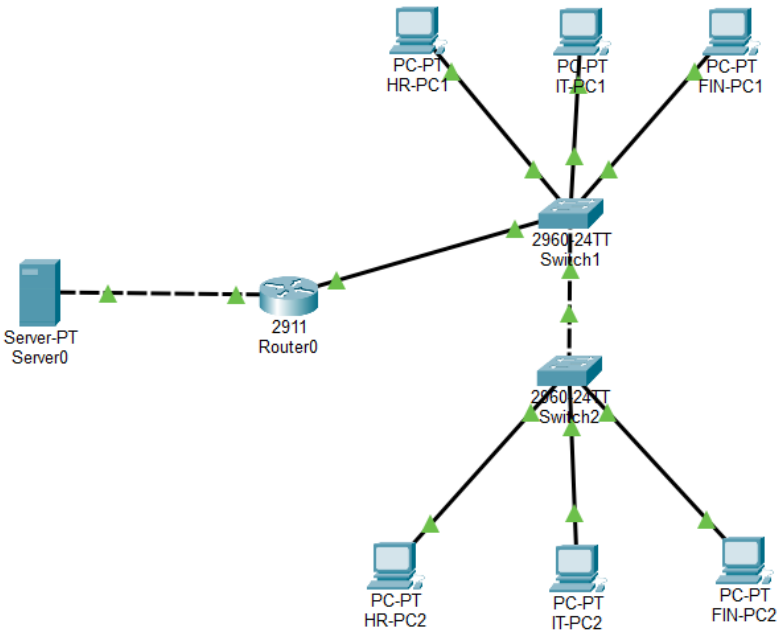


Figure 2: The completed network in Packet Tracer, all links up

2.2 VLANs and subnets

Table 2: VLAN and subnet plan

VLAN	Name	Department	Network	Gateway
10	HR	Human Resources	192.168.10.0/24	192.168.10.1 (G0/0.10)
20	IT	Information Technology	192.168.20.0/24	192.168.20.1 (G0/0.20)
30	FINANCE	Finance	192.168.30.0/24	192.168.30.1 (G0/0.30)
None	Servers	Shared services (routed port)	192.168.40.0/24	192.168.40.1 (G0/1)

2.3 Host addressing

Table 3: Host connections and addresses (all masks 255.255.255.0)

Device	Connected to	VLAN	IP address	Default gateway
HR-PC1	Switch1 Fa0/1	10	192.168.10.11	192.168.10.1
IT-PC1	Switch1 Fa0/2	20	192.168.20.11	192.168.20.1
FIN-PC1	Switch1 Fa0/3	30	192.168.30.11	192.168.30.1
HR-PC2	Switch2 Fa0/1	10	192.168.10.12	192.168.10.1
IT-PC2	Switch2 Fa0/2	20	192.168.20.12	192.168.20.1
FIN-PC2	Switch2 Fa0/3	30	192.168.30.12	192.168.30.1
Server0	Router0 Gi0/1	None	192.168.40.10	192.168.40.1

2.4 Cabling and port map

Table 4: Infrastructure links

Link	Endpoint A	Endpoint B	Cable	Port mode
Switch to switch	Switch1 Fa0/4	Switch2 Fa0/4	Cross-over	802.1Q trunk
Router uplink	Switch1 Fa0/5	Router0 Gi0/0	Straight-through	802.1Q trunk
Server link	Router0 Gi0/1	Server0 Fa0	Cross-over	Routed, untagged
PC links	Fa0/1 to Fa0/3 on each switch	PC FastEthernet0	Straight-through	Access

Trunks are restricted to VLANs 10, 20 and 30. The native VLAN stays at the default (VLAN 1).

3. Implementation Guide

Follow the steps in order. Each step ends with a check, so a fault is found where it was introduced. Commands assume the default hostnames (Switch for each switch, Router for the router); the prompts in the examples are shortened accordingly.

Step 1: Build the topology

- Place one Cisco 2911 router, two Cisco 2960-24TT switches, six PCs and one server.
- Label the PCs HR-PC1, IT-PC1, FIN-PC1, HR-PC2, IT-PC2, FIN-PC2, and the switches Switch1 and Switch2.
- Cable them as in Table 4. Router interfaces are shut down by default, so the router links stay red until Step 5.

Step 2: Create the VLANs and assign access ports

Run this on both switches. The VLAN IDs and port layout are identical on each.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name FINANCE
Switch(config-vlan)# exit
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# end
```

Check: run show vlan brief. VLANs 10, 20 and 30 should be active, with Fa0/1, Fa0/2 and Fa0/3 listed under them.

The VLAN definitions are stored in the VLAN database, not in the running configuration. That is why the exported configuration files in Appendix A do not contain the vlan lines, and why this step must be repeated on any rebuilt switch.

Step 3: Configure the trunks

Switch1

```
Switch(config)# interface fa0/4
Switch(config-if)# description Trunk to Switch2
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# exit
Switch(config)# interface fa0/5
Switch(config-if)# description Trunk to Router G0/0
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# end
```

Switch2

```
Switch(config)# interface fa0/4
Switch(config-if)# description Trunk to Switch 1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# end
```

A 2960 in Packet Tracer supports only 802.1Q, so no encapsulation command is needed (and entering one gives an error). The allowed-VLAN list is optional, but it keeps unused VLANs off the trunks.

Check: run show interfaces trunk. Each trunk port should read trunking, with VLANs 10, 20 and 30 allowed.

Step 4: Address the hosts

On each PC, open Desktop, then IP Configuration, choose Static, and enter the address, mask and default gateway from Table 3. Do the same on Server0 with 192.168.40.10, mask 255.255.255.0 and gateway 192.168.40.1. Screenshots are in Appendix B.

Check: ping from HR-PC1 to HR-PC2 (192.168.10.12). It should succeed, because both are in VLAN 10 and the trunk joins the switches. Pinging a gateway will not work yet.

Step 5: Configure Router-on-a-Stick

```
Router> enable
Router# configure terminal
Router(config)# interface g0/0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface g0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# end
```

- **Enter encapsulation before the IP address.** The router rejects an address on a subinterface that has no VLAN encapsulation yet (see the troubleshooting log).
- **Leave the physical interface without an IP address.** Only the subinterfaces carry addresses.

Check: run show ip interface brief. G0/0.10, G0/0.20 and G0/0.30 should show their gateway addresses as up/up. The router links in the topology turn green. Pings between VLANs now succeed.

Step 6: Configure the server network

```
Router(config)# interface g0/1
Router(config-if)# ip address 192.168.40.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
```

The server uses a routed port instead of a VLAN, so it has no tag. The brief did not specify a server subnet, so 192.168.40.0/24 was chosen to follow the existing pattern.

Step 7: Save the configuration

```
Switch# write memory
Router# write memory
```

Run this on all three devices, then save the Packet Tracer file (Ctrl+S). Without it, the running configuration is lost if a device restarts.

4. Verification of the Build

The commands below confirm each layer of the build before any traffic test. Output for each is shown in the sections that follow.

Table 5: Verification commands and what to look for

Command	What it shows	Observed result
show vlan brief (switches)	VLANs, status and access ports	VLANs 10, 20, 30 active on both switches; trunk ports not listed
show interfaces trunk (switches)	Trunk ports, encapsulation, native and allowed VLANs	Switch1 Fa0/4 and Fa0/5, Switch2 Fa0/4: trunking, 802.1q, VLANs 10, 20, 30
show interfaces status (switches)	Link state, VLAN and speed per port	Fa0/1 to Fa0/3 connected in VLANs 10, 20, 30; trunk ports connected as trunk
show mac address-table (switches)	MAC address learned per VLAN and port	Local PCs on access ports, remote PCs and the router on trunk ports
show ip interface brief (router)	Interface addresses and status	Gi0/0 up, three subinterfaces and Gi0/1 up/up with the planned addresses
show ip route (router)	Routing table	Connected and local routes for all four subnets
show interfaces (router)	MAC address, IP address and VLAN ID per interface	Subinterfaces report 802.1Q VLAN IDs 10, 20 and 30
show running-config (all)	The complete active configuration	Appendix A

4.1 VLANs

Both switches list the three VLANs with one access port each. Fa0/4 (and Fa0/5 on Switch1) are missing from the default list because trunk ports do not belong to a single VLAN.

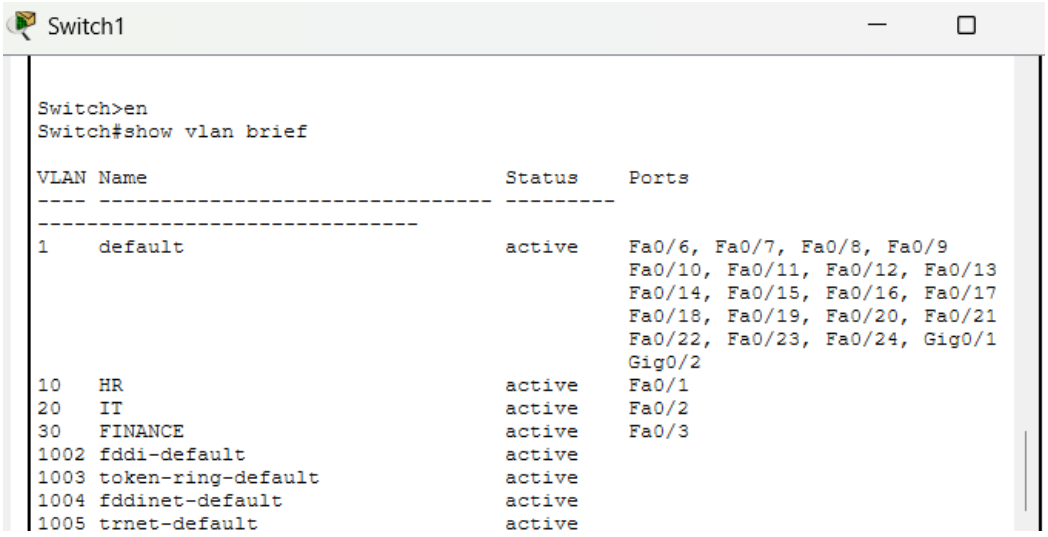


Figure 3: show vlan brief on Switch1

Switch2

```
Switch>en
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
--------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/1
20   IT                               active    Fa0/2
30   FINANCE                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 4: show vlan brief on Switch2

4.2 Trunks

Switch1

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/4       10,20,30
Fa0/5       10,20,30

Port        Vlans allowed and active in management domain
Fa0/4       10,20,30
Fa0/5       10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/4       10,20,30
Fa0/5       10,20,30

Switch#
```

Figure 5: show interfaces trunk on Switch1 (Fa0/4 and Fa0/5)

Switch2

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/4       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/4       10,20,30

Port        Vlans allowed and active in management domain
Fa0/4       10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/4       10,20,30

Switch#
```

Figure 6: show interfaces trunk on Switch2 (Fa0/4)

4.3 Port status

The status output confirms the port roles: access ports show their VLAN, trunk ports show trunk, and all unused ports are notconnect in VLAN 1.

Switch1

```
Switch#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        connected    20         a-full  a-100 10/100BaseTX
Fa0/3                        connected    30         a-full  a-100 10/100BaseTX
Fa0/4     Trunk to Switch2   connected    trunk      a-full  a-100 10/100BaseTX
Fa0/5     Trunk to Router G0 connected    trunk      a-full  a-100 10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
 --More--
```

Figure 7: show interfaces status on Switch1

Switch2

```
Switch#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        connected    20         a-full  a-100 10/100BaseTX
Fa0/3                        connected    30         a-full  a-100 10/100BaseTX
Fa0/4     Trunk to Switch 1  connected    trunk      a-full  a-100 10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
 --More--
```

Figure 8: show interfaces status on Switch2

4.4 MAC address tables

A switch learns a source MAC address on the port the frame arrives on, separately for each VLAN. After traffic was generated, Switch1 held the entries below. The router's address appears under every VLAN on Fa0/5, because its subinterfaces share one MAC address.

Table 6: MAC addresses learned by Switch1

VLAN	MAC address	Port	Device
10	00e0.f9ed.9c1b	Fa0/1	HR-PC1
10	00e0.a3a5.6add	Fa0/4	HR-PC2, reached through Switch2
20	0060.3ea0.b8cd	Fa0/2	IT-PC1

VLAN	MAC address	Port	Device
20	00e0.f908.079a	Fa0/4	IT-PC2, reached through Switch2
30	0002.1625.6e22	Fa0/3	FIN-PC1
30	0040.0b8c.16c1	Fa0/4	FIN-PC2, reached through Switch2
1, 10, 20, 30	0005.5ec8.2401	Fa0/5	Router0 Gi0/0 and its subinterfaces
1, 10, 20, 30	000a.41ca.ae04	Fa0/4	Switch2 (its own port address, seen on the trunk)

Switch2 mirrors this: its own PCs sit on Fa0/1 to Fa0/3, while HR-PC1 and the router are learned on the trunk, Fa0/4. It does not list IT-PC1 or FIN-PC1, because a switch only learns a MAC address from frames that PC itself sends, and in these tests that traffic crossed the router.

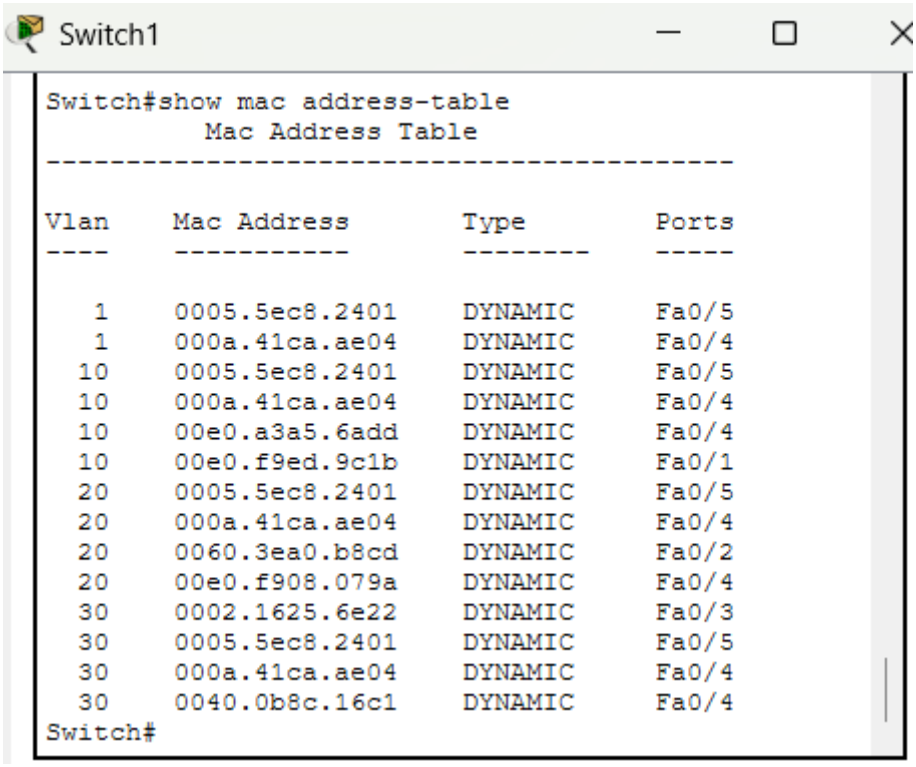


Figure 9: show mac address-table on Switch1

Switch2

```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    00e0.8f12.2c04    DYNAMIC     Fa0/4
  10    00e0.a3a5.6add    DYNAMIC     Fa0/1
  10    00e0.f9ed.9c1b    DYNAMIC     Fa0/4
  20    0005.5ec8.2401    DYNAMIC     Fa0/4
  20    00e0.f908.079a    DYNAMIC     Fa0/2
  30    0005.5ec8.2401    DYNAMIC     Fa0/4
  30    0040.0b8c.16c1    DYNAMIC     Fa0/3
Switch#
```

Figure 10: show mac address-table on Switch2

4.5 Router

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     192.168.40.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down
down
Vlan1                  unassigned      YES unset  administratively down
down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/1

Router#
```

Figure 11: show ip interface brief and show ip route on Router0

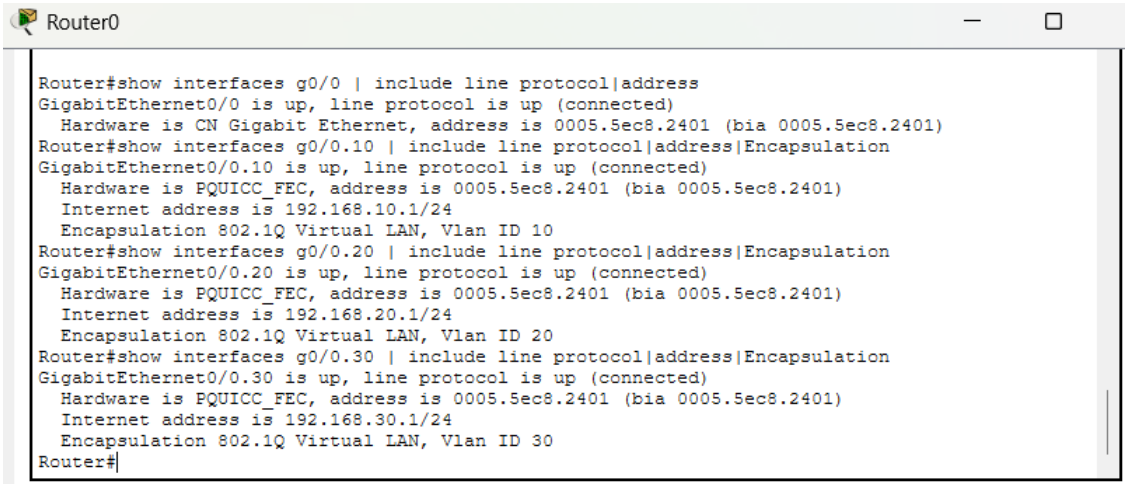


Figure 12: show interfaces (filtered to status, MAC, IP and encapsulation lines)

All four interfaces share MAC address 0005.5ec8.2401, and each subinterface reports its own 802.1Q VLAN ID. G0/1, the server port, is a separate physical interface and is not part of the trunk.

5. Functional Testing

5.1 Test A: same-VLAN communication

Each department's first PC pinged its second PC across the Switch1-Switch2 trunk. All replies arrived with TTL 128, which means no router was involved.

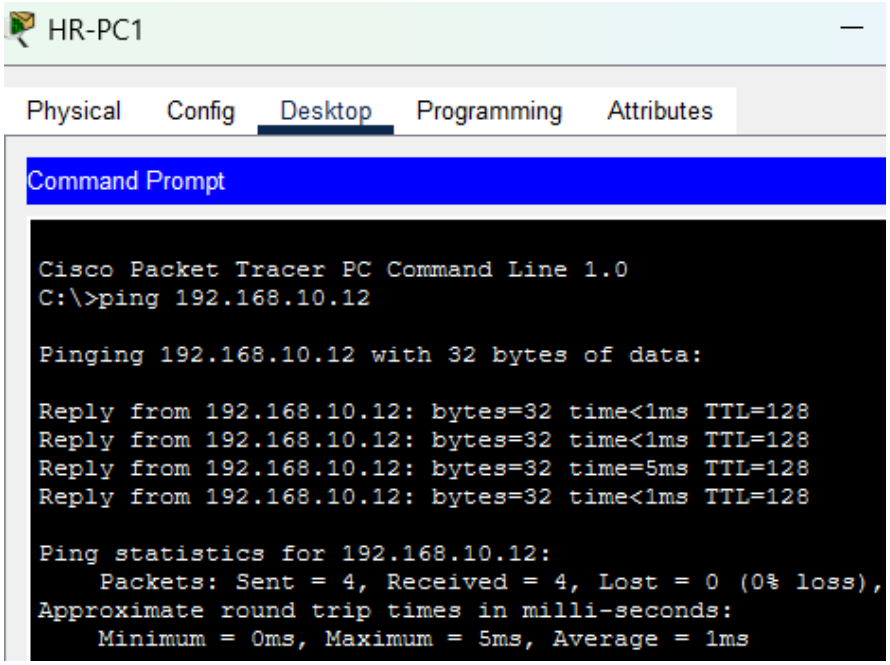


Figure 13: HR-PC1 to HR-PC2

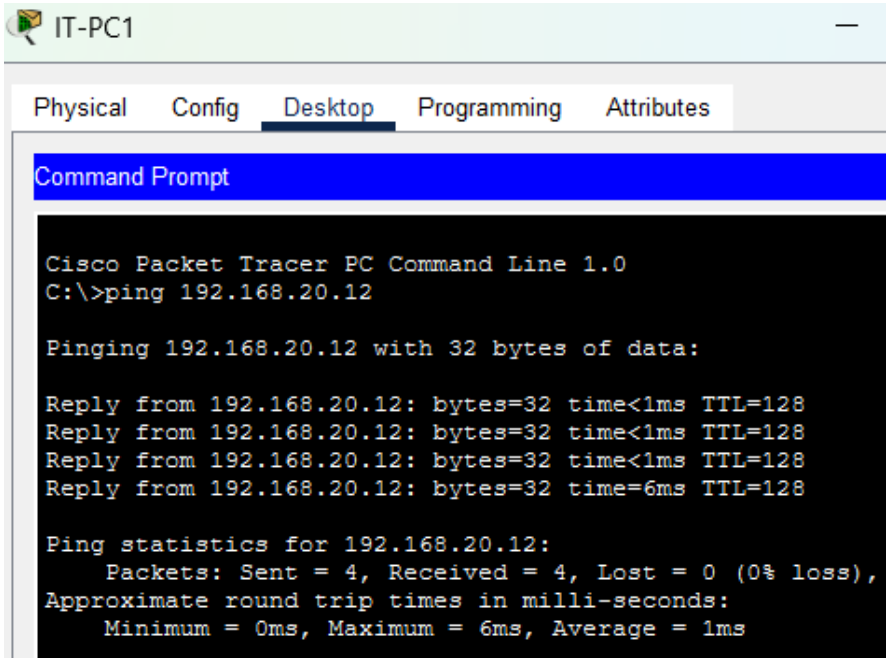


Figure 14: IT-PC1 to IT-PC2

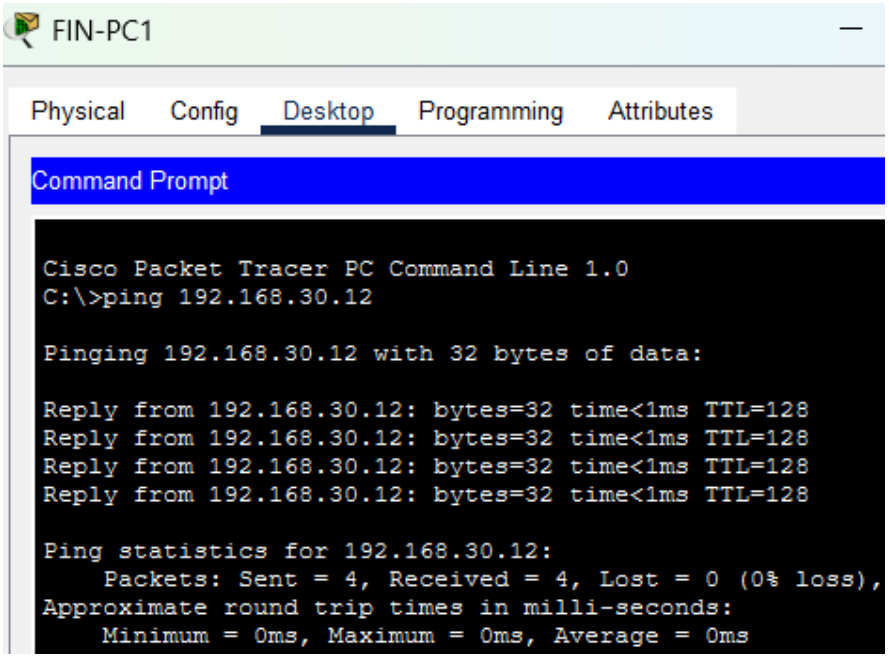


Figure 15: FIN-PC1 to FIN-PC2

5.2 Test B: inter-VLAN communication

Pings between departments went through the router. Replies show TTL 127, one less than the PC's starting value of 128, because the router decrements it. The first request in two tests timed out, which is the ARP lookup for the gateway, and the repeat ping succeeded.

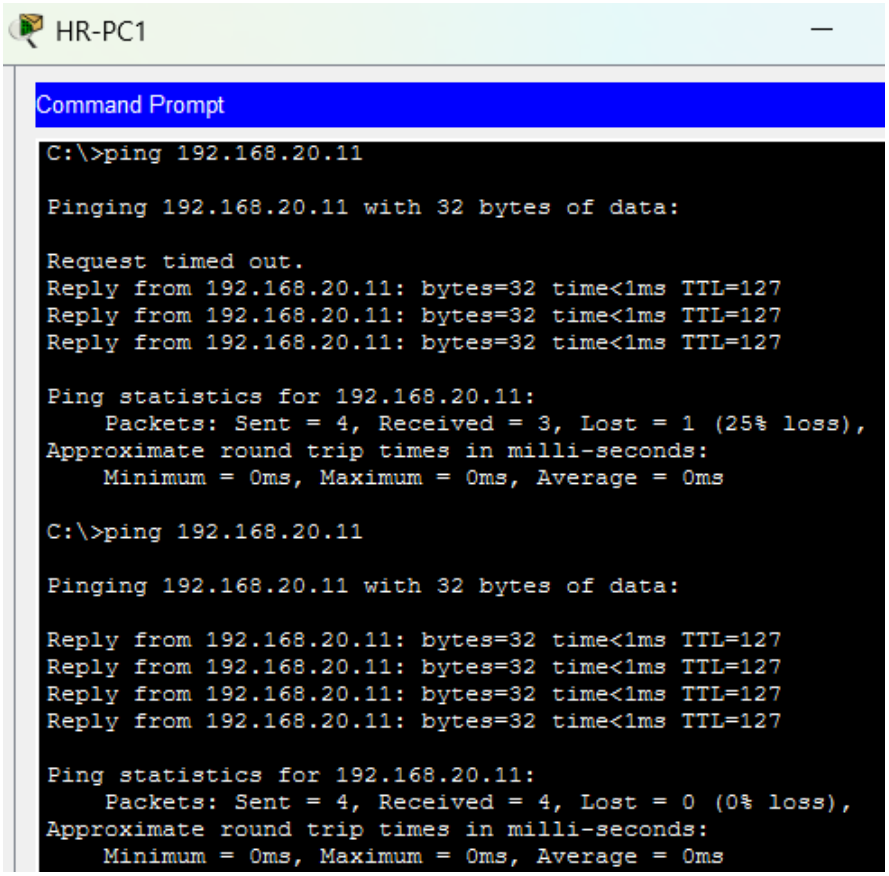


Figure 16: HR-PC1 to IT-PC1 (first request lost to ARP, then 4 of 4)

HR-PC1

Command Prompt

```
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time=1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time=4ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

Figure 17: HR-PC1 to FIN-PC1 (first request lost to ARP, then 4 of 4)

FIN-PC1

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=127
Reply from 192.168.10.11: bytes=32 time<1ms TTL=127
Reply from 192.168.10.11: bytes=32 time<1ms TTL=127
Reply from 192.168.10.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 18: FIN-PC1 to HR-PC1 (4 of 4)

5.3 Test C: behaviour when the router link fails

The router's Gi0/0 interface was shut down with the shutdown command. The topology shows the link red, and the interface table shows Gi0/0 and its three subinterfaces administratively down. The server port, G0/1, stays up because it is independent of the trunk.

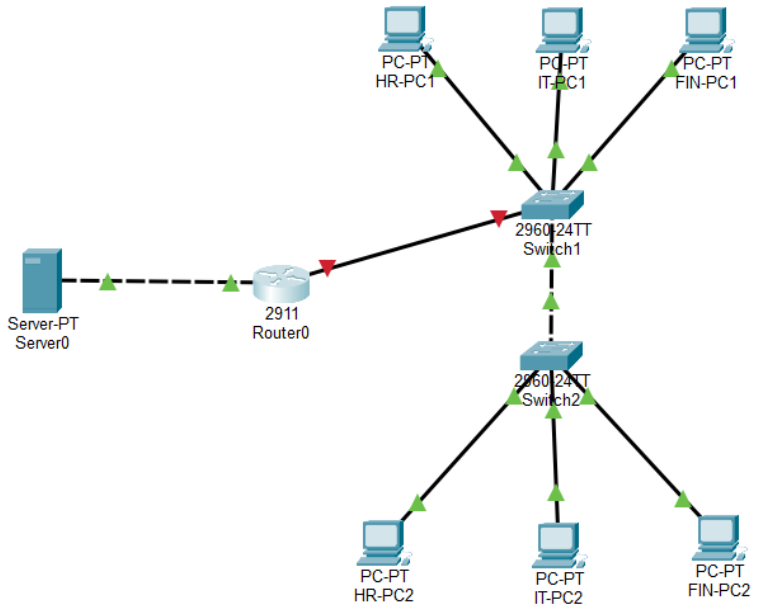


Figure 19: Router link down: Router0 to Switch1 shown red

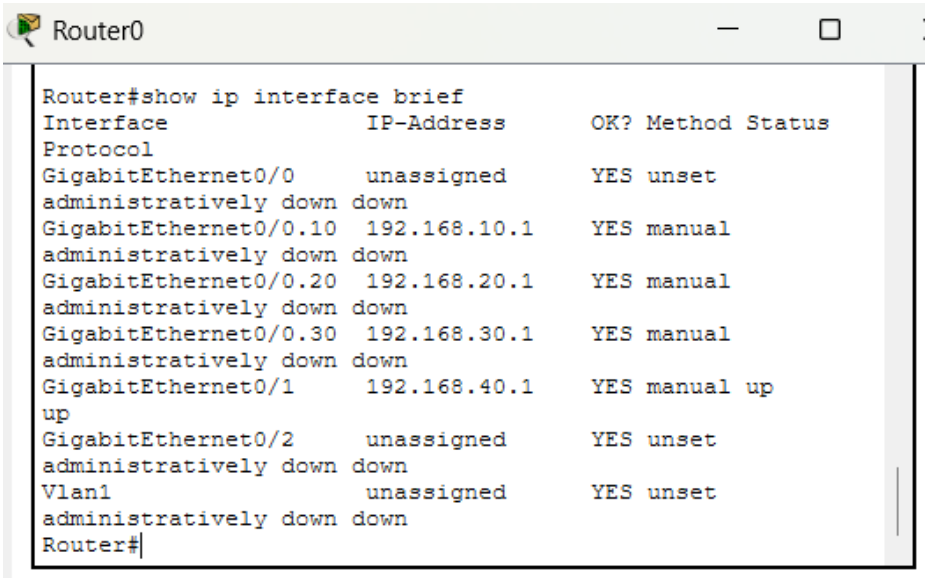


Figure 20: show ip interface brief with Gi0/0 shut down

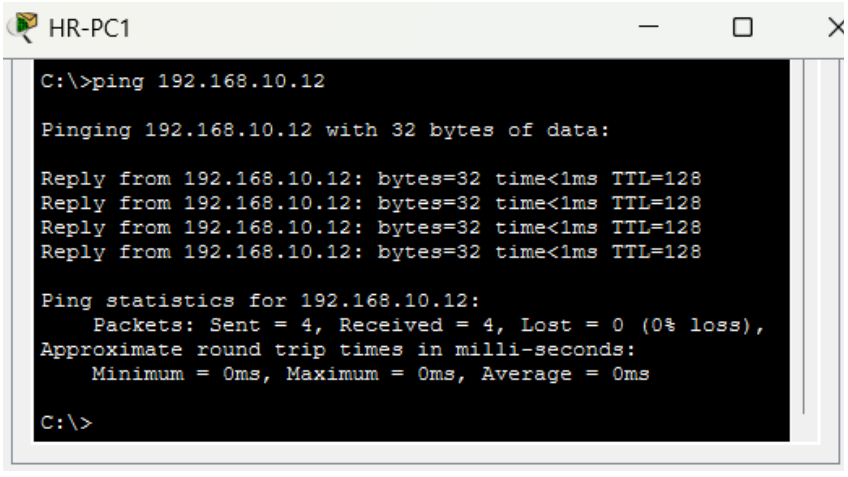


Figure 21: Same-VLAN ping still succeeds (HR-PC1 to HR-PC2, TTL 128)

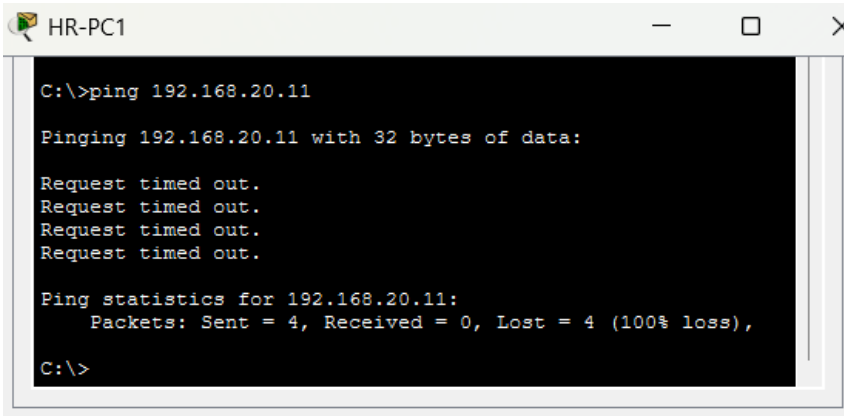


Figure 22: Inter-VLAN ping fails (HR-PC1 to IT-PC1, 100% loss)

The interface was then restored with no shutdown. The first restored ping lost two requests while the link and ARP came back, and the repeat succeeded.

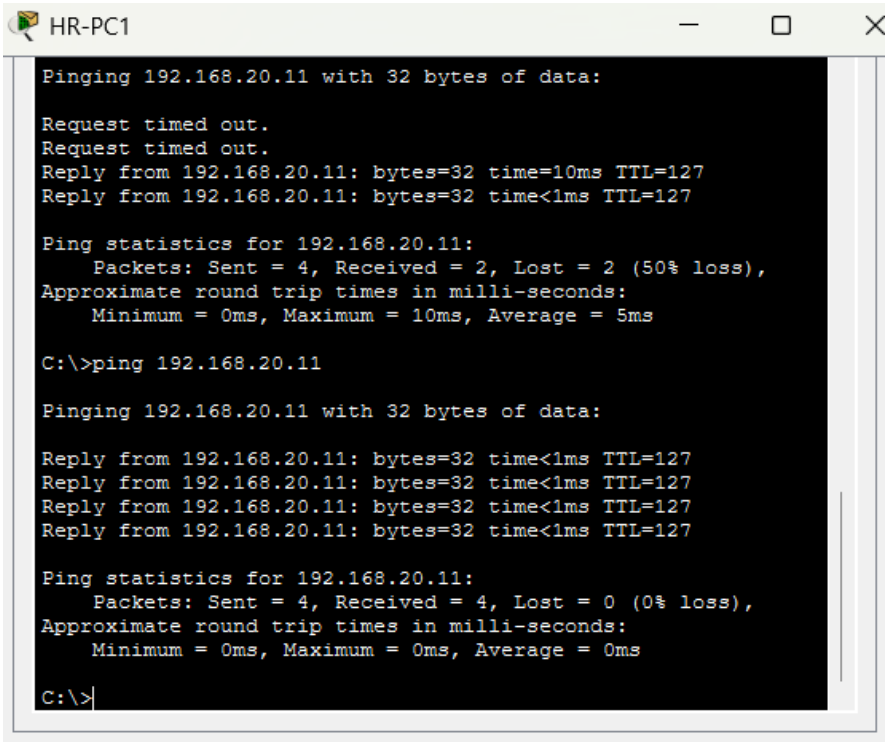


Figure 23: Inter-VLAN ping after restoring Gi0/0

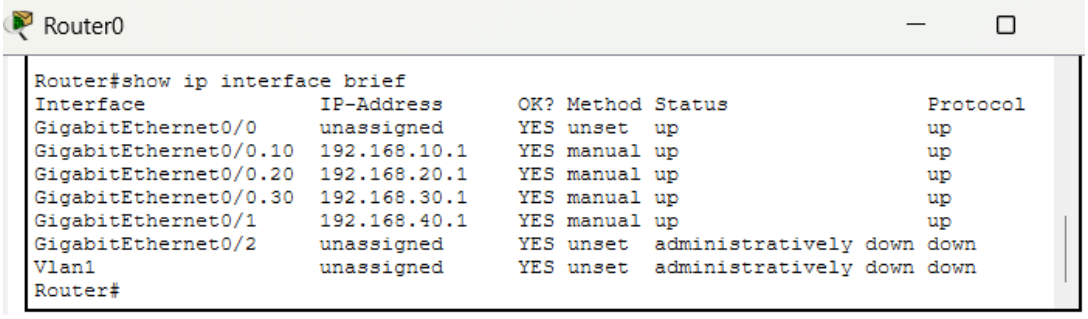


Figure 24: show ip interface brief after restoring Gi0/0, all interfaces up/up

Questions

- **Can two PCs in the same VLAN still communicate?** Yes. The switches forward their frames at Layer 2 using MAC addresses over the trunk, and the router is not on that path.
- **Can a PC in VLAN 10 communicate with a PC in VLAN 20?** No. The packet has to go to the router, and with the link down it cannot.
- **Why does the router affect inter-VLAN traffic but not same-VLAN traffic?** Devices in one VLAN share a network, and the switches carry their frames by MAC address. Different VLANs are separate networks, so traffic between them must be sent to the default gateway, which is the router's subinterface, and routed at Layer 3 by IP address. With the router link down the gateway is unreachable, so inter-VLAN communication fails while same-VLAN communication continues.

5.4 Test D: packet-flow analysis in Simulation Mode

Simulation Mode was used with ARP and ICMP filters and a single ping from HR-PC1 to FIN-PC1. Two captures were taken. Run 1 started with empty ARP caches, so ARP and ICMP both appear. Run 2 was ICMP only, because the caches were already filled. Full-size captures are in Appendix B.

Run 1 event sequence (request path)

Table 7: Simulation events for the ping from HR-PC1 to FIN-PC1

Time (s)	Movement	What happens
0.001	HR-PC1 to Switch1	ARP request for the gateway, 192.168.10.1
0.002	Switch1 to Switch2 and Router0	Broadcast flooded within VLAN 10 only
0.003	Router0 to Switch1	Router answers with its MAC address
0.004	Switch1 to HR-PC1	HR-PC1 learns the gateway MAC; the queued ICMP is released
0.005 to 0.006	HR-PC1 to Switch1 to Router0	Echo request sent to the router's MAC
0.007	Router0 to Switch1	Router forwards the packet toward VLAN 30
0.008	Switch1 to FIN-PC1	Echo request delivered
0.008 to 0.012	FIN-PC1, Switch1, Router0	FIN-PC1 ARPs for its own gateway, 192.168.30.1, so it can reply
0.013 to 0.016	FIN-PC1 back to HR-PC1	Echo reply returns through Switch1, Router0 and Switch1

No ARP request from the router appears in the list, so its ARP table already held FIN-PC1's address from earlier tests. The ARP broadcast from HR-PC1 reached only HR-PC2 and the router, and FIN-PC1's broadcast reached only FIN-PC2 and the router. That is VLAN broadcast isolation in practice.

Q1. How does the source PC learn its gateway's MAC address?

HR-PC1 compares 192.168.30.11 with its own network, 192.168.10.0/24, and sees it is remote, so the frame must go to the default gateway, 192.168.10.1. It does not know that address's MAC, so it sends an ARP request, a broadcast to FFFF.FFFF.FFFF. Switch1 floods it within VLAN 10, and Router subinterface G0/0.10 answers with a unicast ARP reply carrying 0005.5EC8.2401. HR-PC1 stores the answer and uses it as the destination MAC of the ICMP frame (Figures B.1 and B.2).

Q2. How does the frame travel from the access switch to the router?

HR-PC1 sends an ordinary untagged Ethernet frame into access port Fa0/1. Switch1 knows that port belongs to VLAN 10, so when it forwards the frame out the Fa0/5 trunk it adds an 802.1Q tag (TPID 0x8100, TCI 0x000a, which is VLAN 10). Router0 receives the tagged frame on Gi0/0 and hands it to the matching subinterface, G0/0.10 (Figures B.3 and B.4).

Q3. How does the router choose the outgoing interface?

The VLAN tag tells the router which subinterface the packet arrived on. It then reads the destination IP address, 192.168.30.11, and finds it in the directly connected route 192.168.30.0/24 on G0/0.30. It looks up FIN-PC1's MAC address in its ARP table, builds a new frame tagged for VLAN 30 (TCI 0x001e), and sends it out the same physical port (Figure B.5 and the routing table in Section 4.5).

Q4. What happens to the MAC addresses when the router forwards the packet?

Both Ethernet addresses are replaced at the router, because each hop builds a new Layer 2 frame.

Table 8: Ethernet and IP header values along the request path

Point	Source MAC	Destination MAC	VLAN tag	TTL
HR-PC1 sends	00E0.F9ED.9C1B	0005.5EC8.2401	None	255
Router0 receives	00E0.F9ED.9C1B	0005.5EC8.2401	VLAN 10	255
Router0 sends	0005.5EC8.2401	0002.1625.6E22	VLAN 30	254
FIN-PC1 receives	0005.5EC8.2401	0002.1625.6E22	None	254

The source and destination IP addresses are 192.168.10.11 and 192.168.30.11 in every row. The tag is added by Switch1 on the trunk and removed on the access port toward FIN-PC1 (Figures B.4 to B.7). The router uses the same MAC address on both sides because its subinterfaces share the physical port's address.

Q5. Do the source and destination IP addresses change?

No. Without NAT, routing leaves both IP addresses unchanged from end to end. Only the Ethernet addresses are rewritten and the TTL is reduced by one at the router. The reply (Figure B.8) swaps them, with source 192.168.30.11 and destination 192.168.10.11, and arrives at HR-PC1 with TTL 127, addressed from the router's MAC (Figure B.9).

5.5 Test E: server reachability

Each VLAN reaches the server through the router with TTL 127. HR-PC1 lost its first request to ARP and succeeded on the repeat. IT-PC1, FIN-PC1 and FIN-PC2 also reached the server (Figure B.10), and FIN-PC2 shows that traffic from the second switch crosses both trunks.

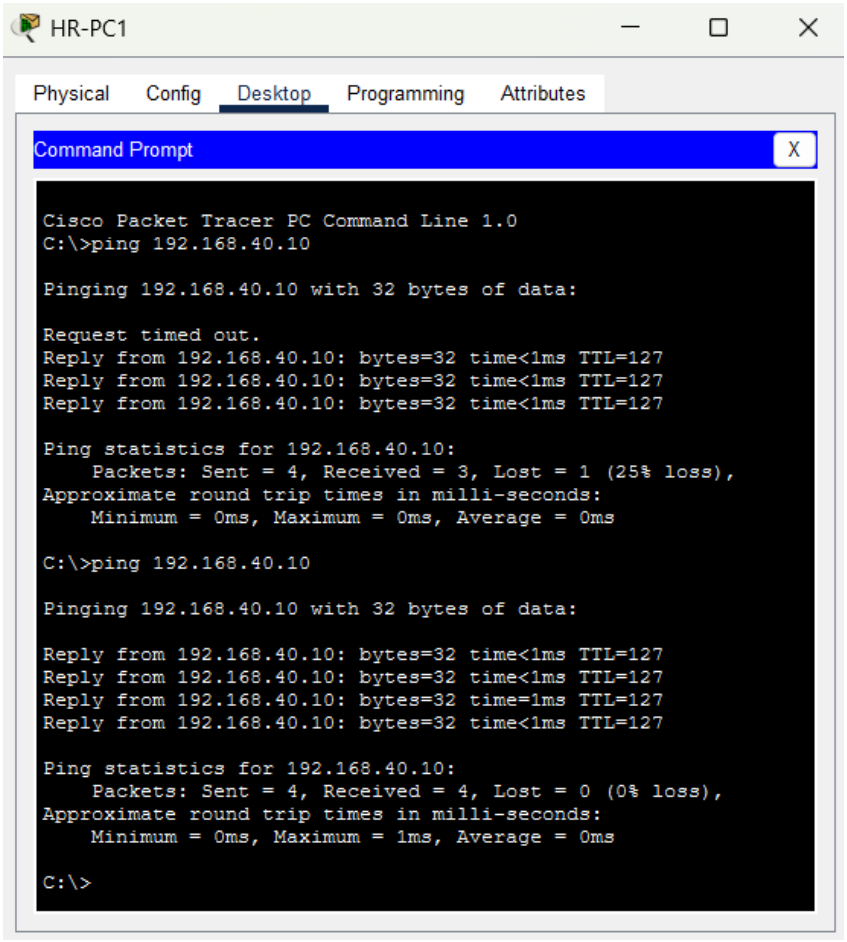


Figure 25: HR-PC1 to Server0 (192.168.40.10)

6. Troubleshooting Log

Each fault below was met during the build. The table gives the symptom, the cause and the fix. Evidence for the main faults follows the table.

Table 9: Faults, causes and fixes

#	Symptom	Cause	Resolution
1	Router link red; Switch1 Fa0/5 notconnect although set as a trunk	Router ports are shut down by default, so the far end of the link was down	no shutdown on Router0 Gi0/0; the link turned green
2	ip address on G0/0.20 rejected with a message about 802.1Q	The address was entered before the subinterface had an encapsulation	Set encapsulation dot1Q first, then the address
3	G0/0.20 given encapsulation dot1Q 30 by mistake	Typing error	Re-entered encapsulation dot1Q 20, which replaces the old value; confirmed with show interfaces
4	HR-PC1 could not ping FIN-PC1: 100% loss	The gateway addresses did not exist yet because the subinterfaces were not finished	Completed Router-on-a-Stick; Test B then passed
5	First ping of an inter-VLAN test times out	Expected: the first packet waits for ARP to resolve the gateway	None needed; the repeat ping succeeds
6	An empty router rip process in the router configuration	Entered unintentionally; it had no network statements, so it had no effect	Removed with no router rip, saved, and the final configuration re-exported
7	MAC address tables showed no PC entries	Dynamic entries had aged out after a period without traffic	Generated fresh pings and re-ran the command at once
8	Simulation Mode reported Buffer Full	The event buffer filled during capture	Cleared the event list, reset the simulation, and used a single Simple PDU stepped with Capture/Forward

Fault 1: link down before the router interface was enabled

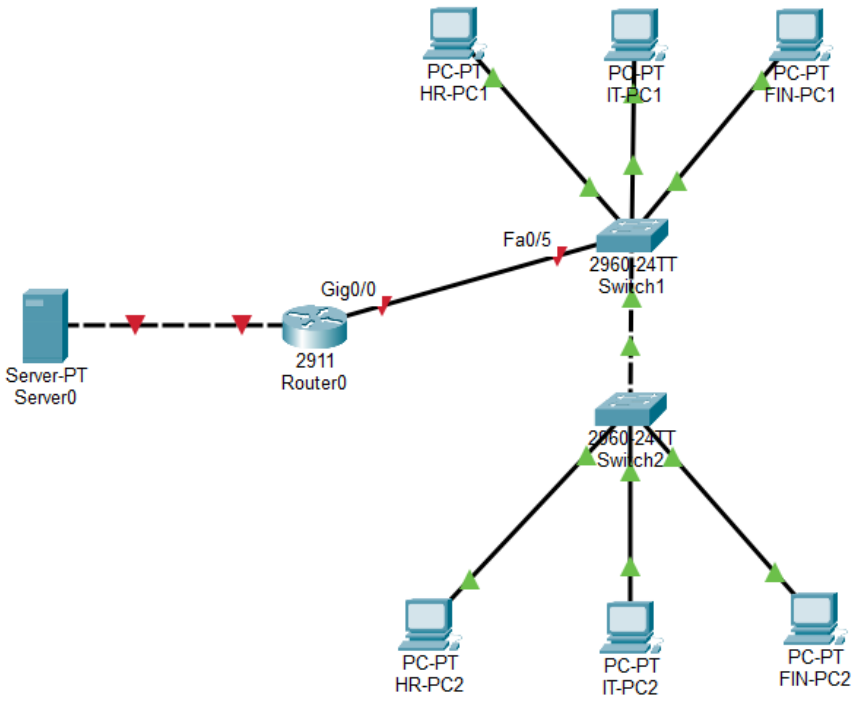


Figure 26: Router0 Gig0/0 to Switch1 Fa0/5 shown red

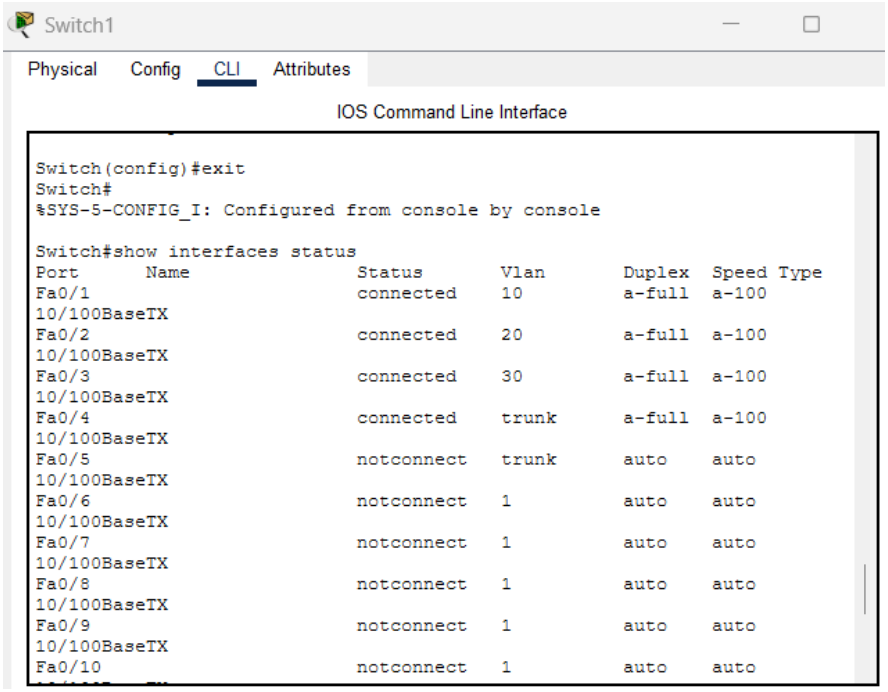


Figure 27: Switch1 Fa0/5 configured as a trunk but notconnect

Faults 2 and 3: subinterface rejected, then wrong encapsulation

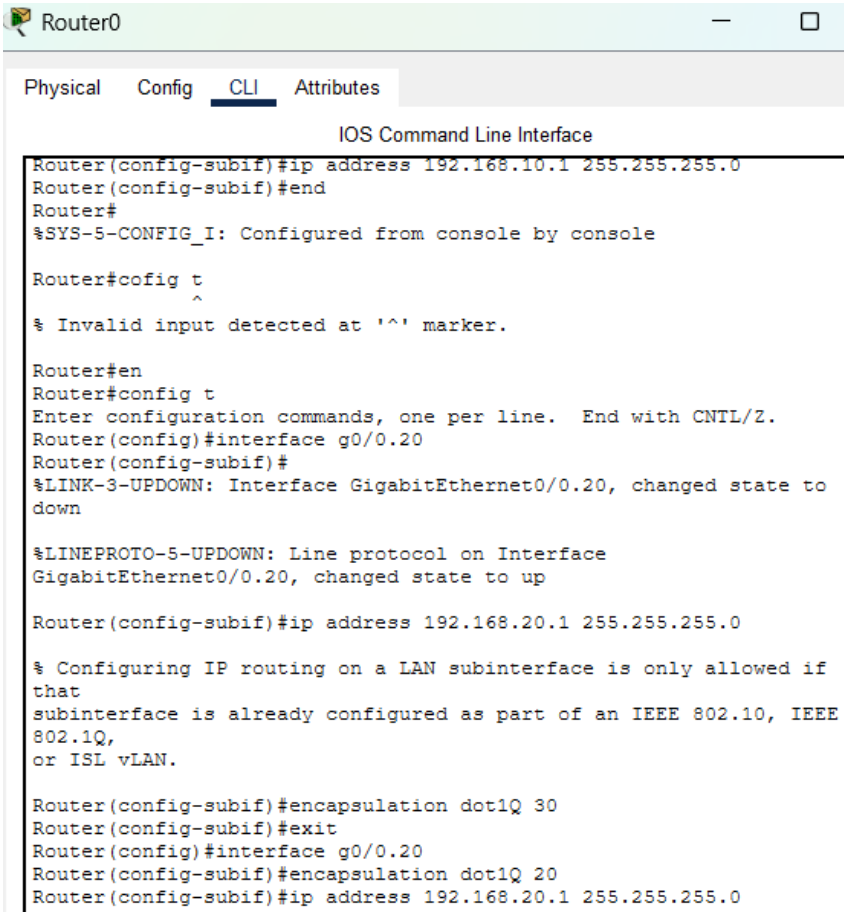


Figure 28: The rejected ip address, the dot1Q 30 typo, and the corrected dot1Q 20

Fault 4: no inter-VLAN connectivity before routing was configured

The failed ping, with HR-PC1 and the FIN-PC1 address details, is reproduced at full size as Figure B.11 in Appendix B.

Fault 6: the empty RIP process

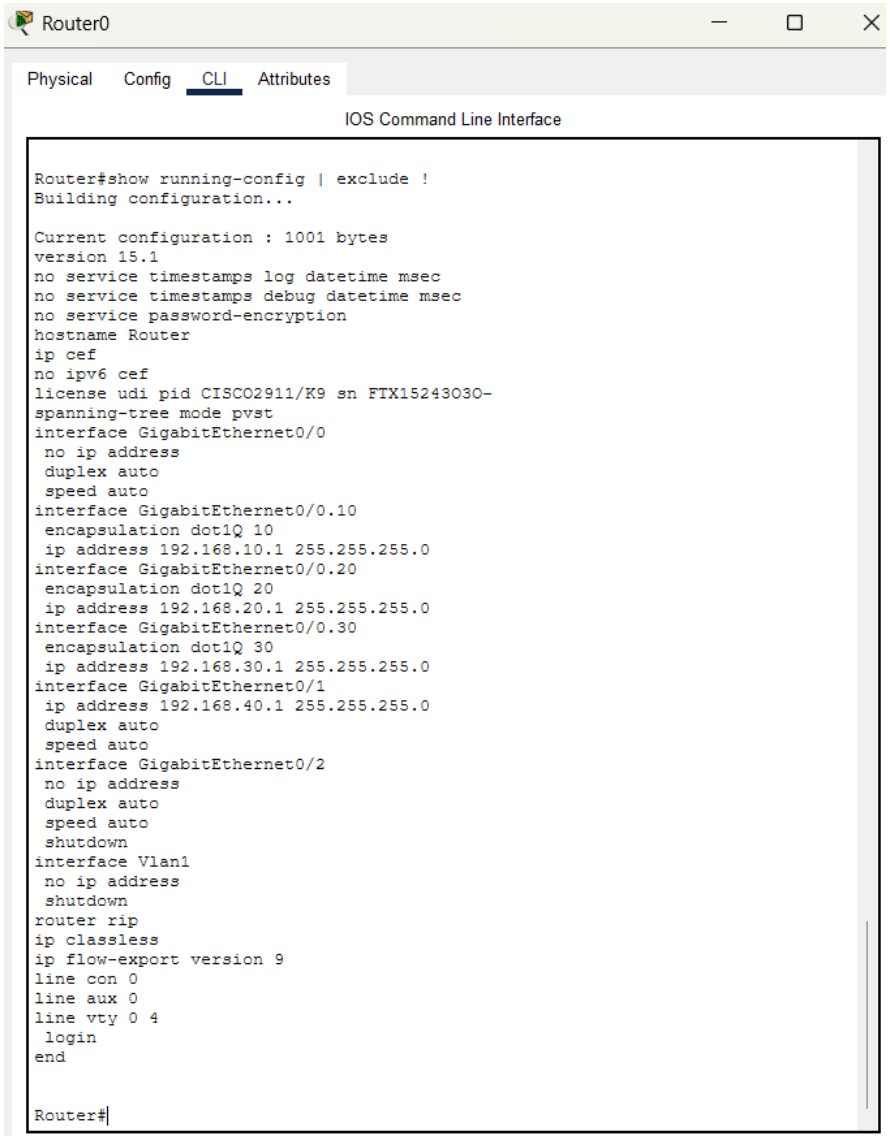


Figure 29: show running-config with the unintended router rip line

Fault 7: empty MAC tables

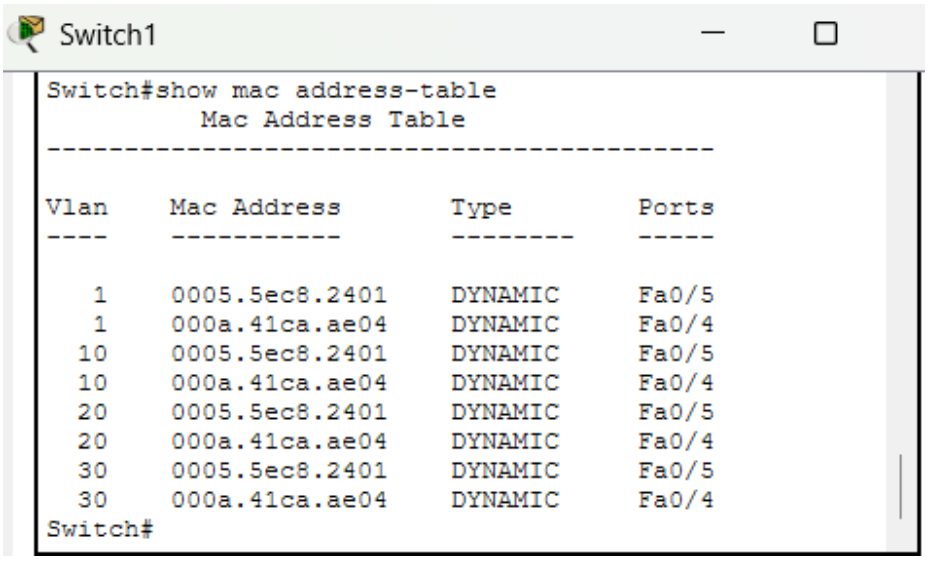


Figure 30: Switch1 MAC table before fresh traffic: no PC entries

Fault 8: simulation buffer full

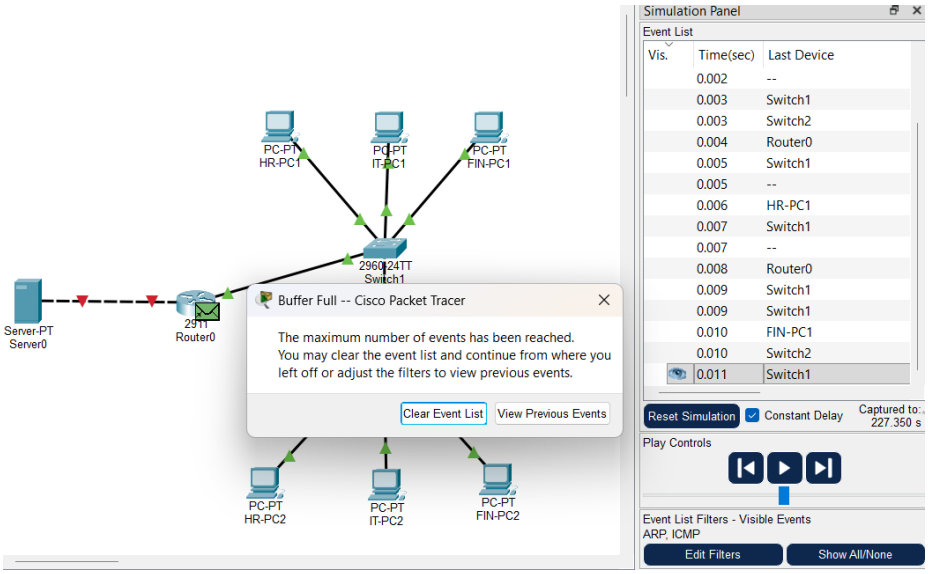


Figure 31: Packet Tracer Buffer Full message

7. Conclusion

The project met its objectives. Three departments now sit in separate VLANs and subnets, the departments are spread across two switches, and a single router link provides all inter-VLAN routing.

How VLANs improve organisation and isolation

- **Logical grouping regardless of location.** HR-PC1 and HR-PC2 sit on different switches yet behave as one network, because the trunk carries VLAN 10 between them.
- **Smaller broadcast domains.** In the packet trace, the ARP broadcast from HR-PC1 reached only the HR PC and the router, and the Finance broadcast reached only the Finance PC and the router. IT devices saw neither.
- **A single control point between departments.** Every packet between VLANs must pass through the router. The failure test proved this: with the router link down, departments were fully isolated from each other while each continued to work internally.
- **Simple growth.** A new department needs a VLAN, a subnet and one subinterface, not new cabling. The server was added in the same way, as one more routed network.

VLANs separate the networks, but this design routes freely between them. Restricting which departments may reach each other is the next step, using access control lists on the router. Other sensible hardening steps are DHCP for addressing, port security on access ports, shutting unused ports, and moving the native VLAN away from VLAN 1.

Acknowledgement

This project is adapted from a VLAN and inter-VLAN routing exercise in Devtown's "Networking Basics for Cybersecurity" course material. The addressing, configuration, testing and documentation here were carried out and extended independently, including the shared server, the verification of MAC address tables and the packet-level analysis.

Appendix A: Final Configurations

The listings show the final saved configuration of each device, with empty interface sections and blank separator lines removed for readability. The complete exports are kept unaltered in the project repository. To rebuild a switch, apply the VLAN lines first, because the VLAN database does not appear in the running configuration.

A.1 Switch1

VLAN database (entered in Step 2, not shown by show running-config)

```
vlan 10
 name HR
vlan 20
 name IT
vlan 30
 name FINANCE
```

Running configuration

```
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname Switch
spanning-tree mode pvst
spanning-tree extend system-id
```

```
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/4
 description Trunk to Switch2
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
interface FastEthernet0/5
 description Trunk to Router G0/0
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
interface Vlan1
 no ip address
 shutdown
line vty 0 4
 login
line vty 5 15
 login
end
```

A.2 Switch2

VLAN database

```
vlan 10
 name HR
vlan 20
 name IT
vlan 30
 name FINANCE
```

Running configuration

```
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname Switch
spanning-tree mode pvst
spanning-tree extend system-id
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
interface FastEthernet0/2
 switchport access vlan 20
 switchport mode access
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/4
```

```
 description Trunk to Switch 1
 switchport trunk allowed vlan 10,20,30
 switchport mode trunk
interface Vlan1
 no ip address
 shutdown
line vty 0 4
 login
line vty 5 15
 login
end
```

A.3 Router0

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname Router
ip cef
no ipv6 cef
spanning-tree mode pvst
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
interface GigabitEthernet0/1
 ip address 192.168.40.1 255.255.255.0
 duplex auto
 speed auto
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
interface Vlan1
 no ip address
 shutdown
ip classless
ip flow-export version 9
line vty 0 4
 login
end
```

Appendix B: Evidence at Full Size

Screenshots are reproduced exactly as captured. B.1 to B.9 are the Simulation Mode captures for Test D, in the order they are cited in Section 5.4; B.10 and B.11 are referred to in Sections 5.5 and 6; B.12 to B.14 show the host addressing. Figures B.1 and B.2 and B.4 to B.8 come from Run 1, and B.3 and B.9 from Run 2. The PDU tabs named below are the tabs open in each capture.

B.1 Simulation captures (Test D)

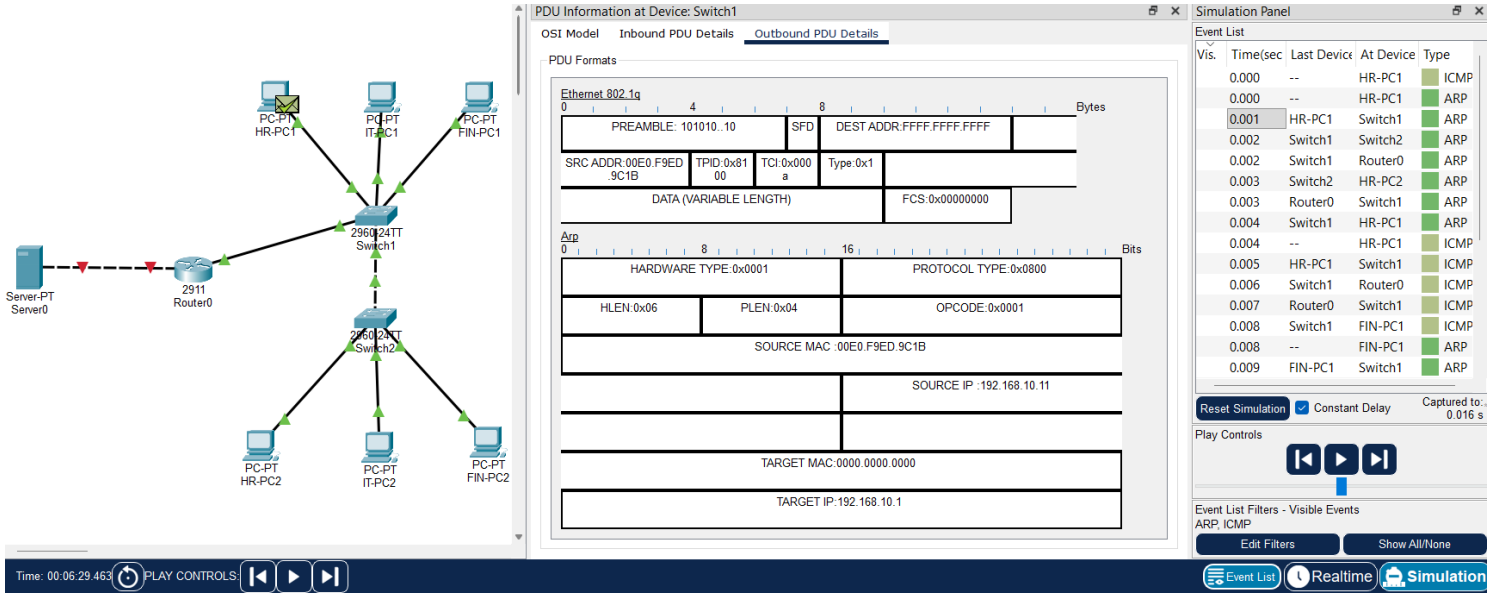


Figure B.1: Switch1, outbound PDU: ARP request for 192.168.10.1, broadcast destination FFFF.FFFF.FFFF, tagged VLAN 10

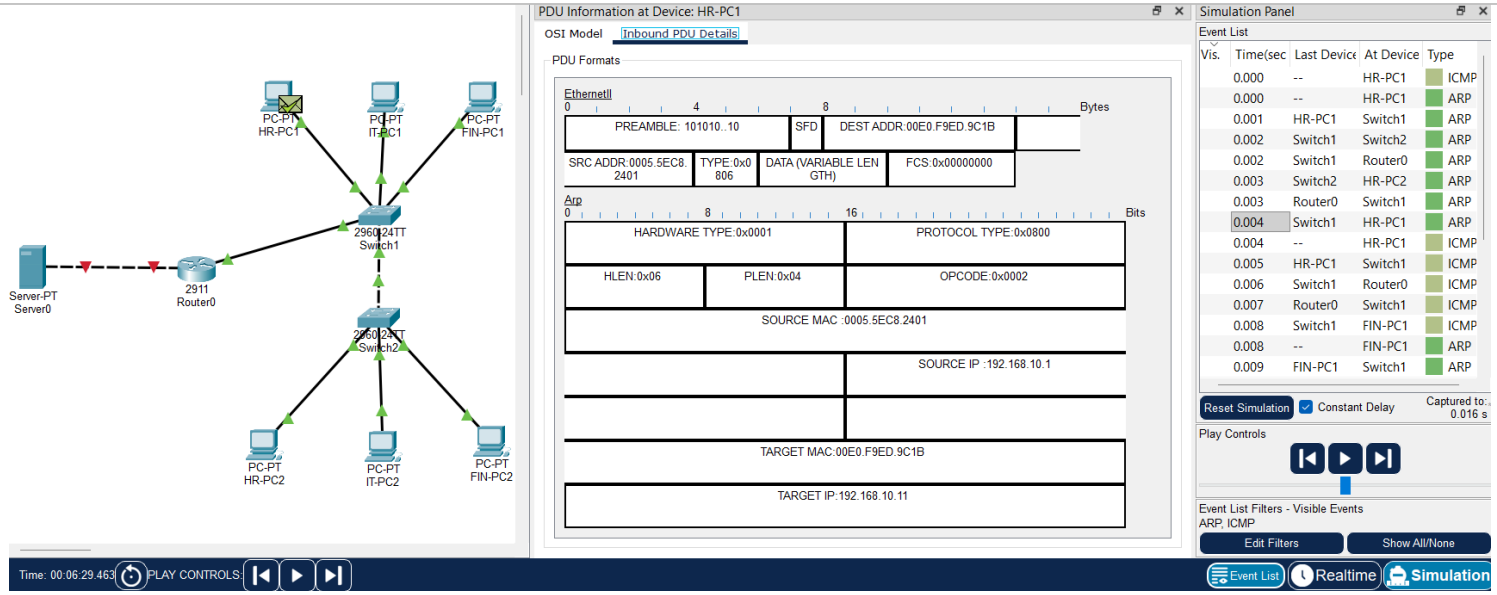


Figure B.2: HR-PC1, inbound PDU: ARP reply carrying the router's MAC address 0005.5EC8.2401

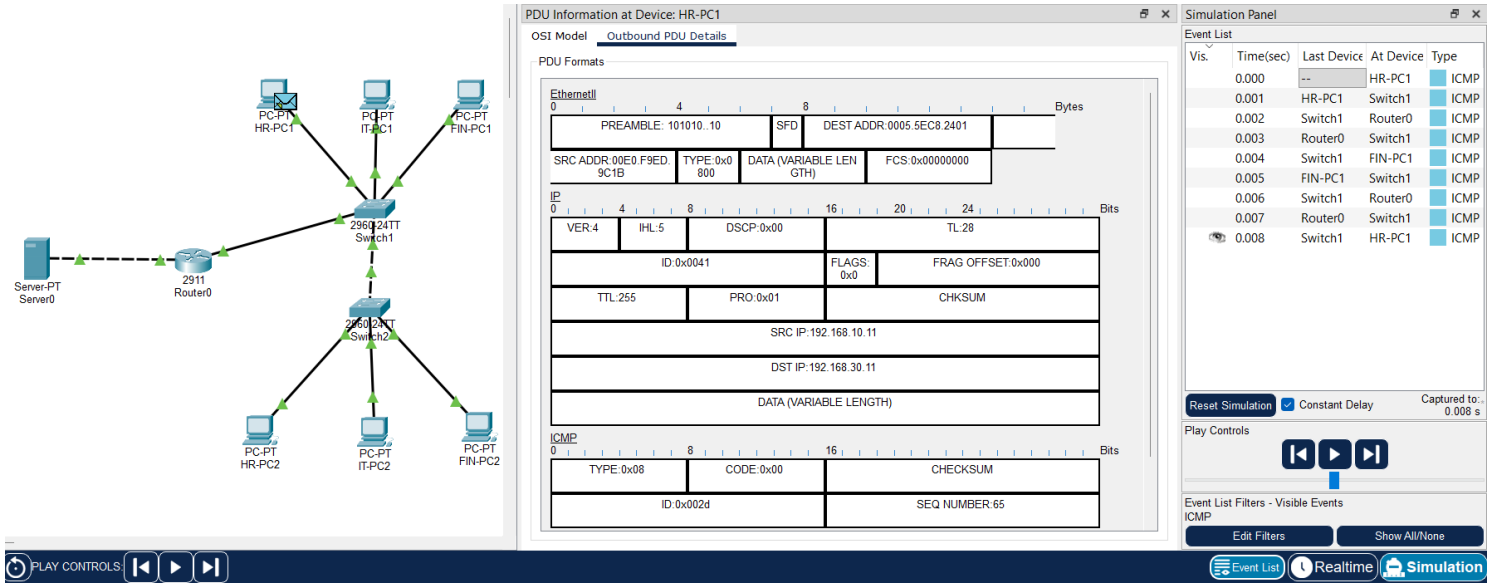


Figure B.3: HR-PC1, outbound PDU (Run 2): echo request addressed to the router's MAC, IP 192.168.10.11 to 192.168.30.11

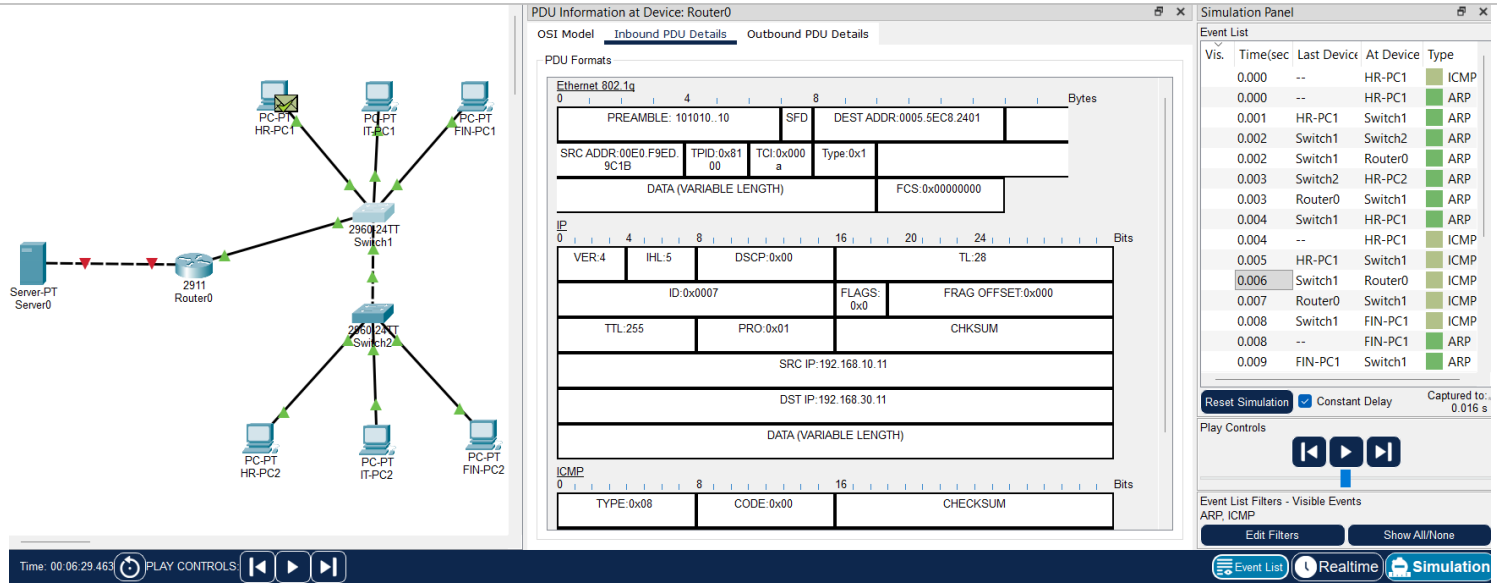


Figure B.4: Router0, inbound PDU: 802.1Q frame tagged VLAN 10 (TCI 0x000a), TTL 255

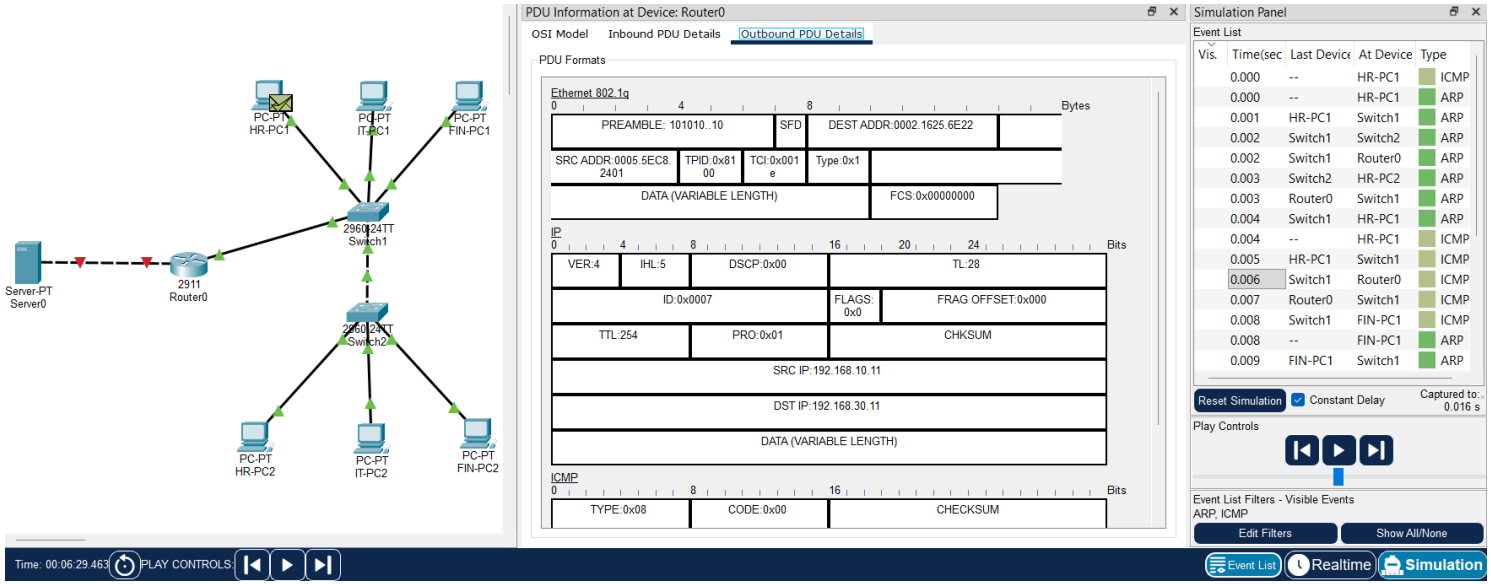


Figure B.5: Router0, outbound PDU: MACs rewritten, tagged VLAN 30 (TCI 0x001e), TTL 254

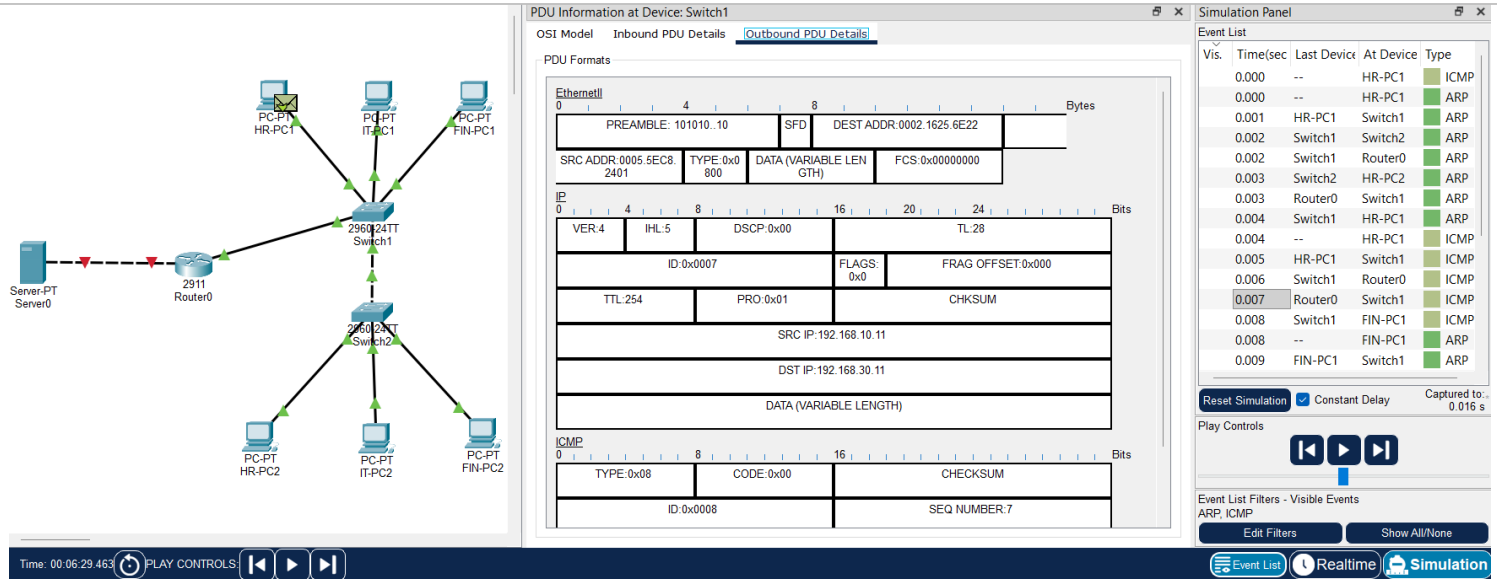


Figure B.6: Switch1, outbound PDU toward FIN-PC1: untagged Ethernet II frame

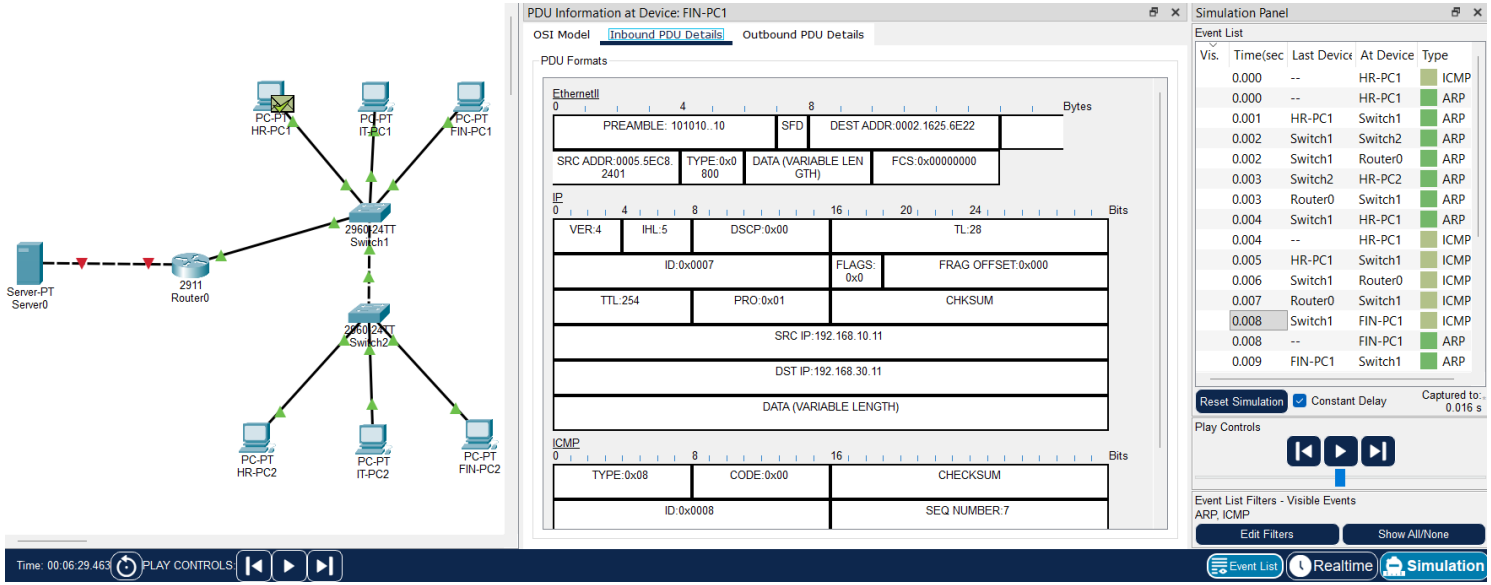


Figure B.7: FIN-PC1, inbound PDU: echo request, IP addresses unchanged

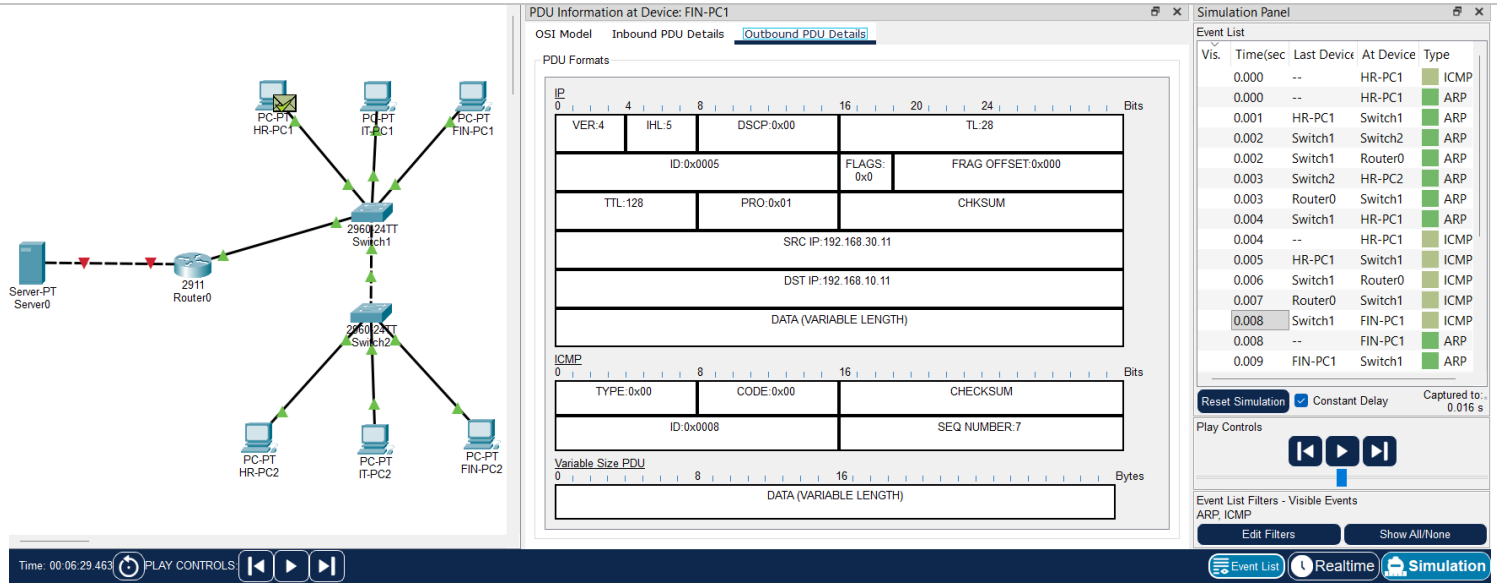


Figure B.8: FIN-PC1, outbound PDU: echo reply, source 192.168.30.11, destination 192.168.10.11

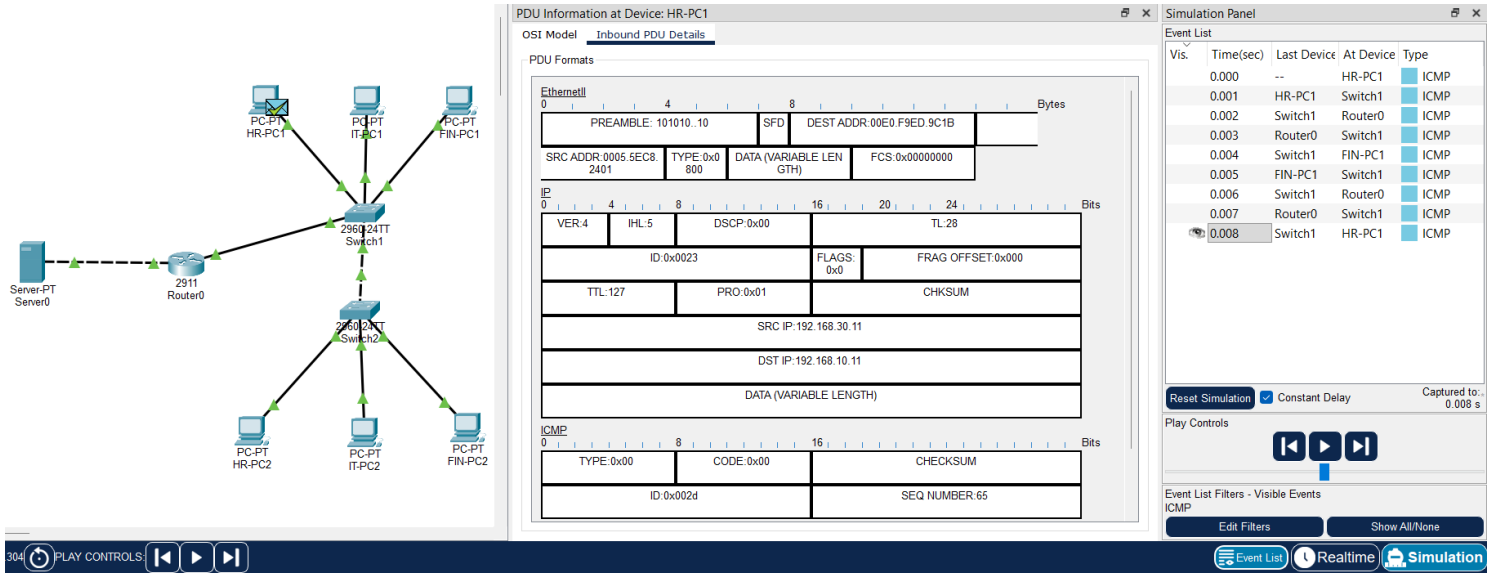


Figure B.9: HR-PC1, inbound PDU (Run 2): echo reply from the router's MAC, TTL 127

B.2 Other evidence

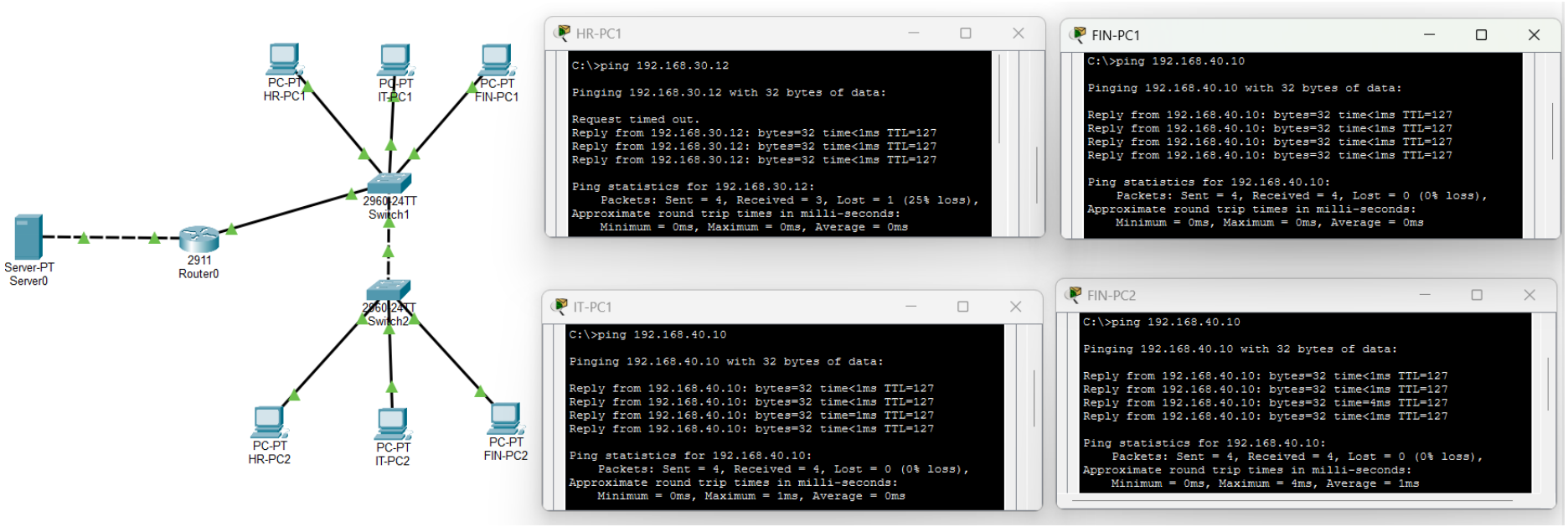


Figure B.10: Server0 reachable from IT-PC1, FIN-PC1 and FIN-PC2 (TTL 127), with an inter-VLAN ping from HR-PC1 to FIN-PC2

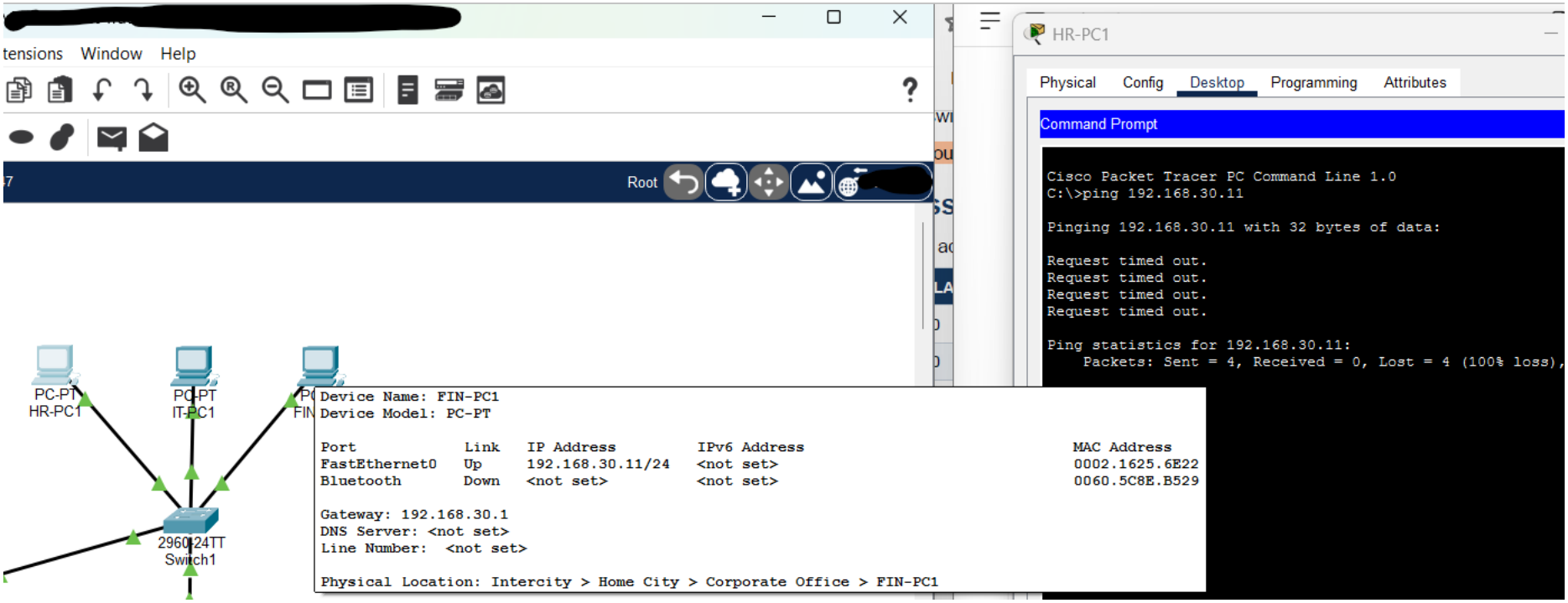


Figure B.11: Fault 4: HR-PC1 pinging FIN-PC1 with 100% loss before the router subinterfaces were configured

B.3 Host configuration screens

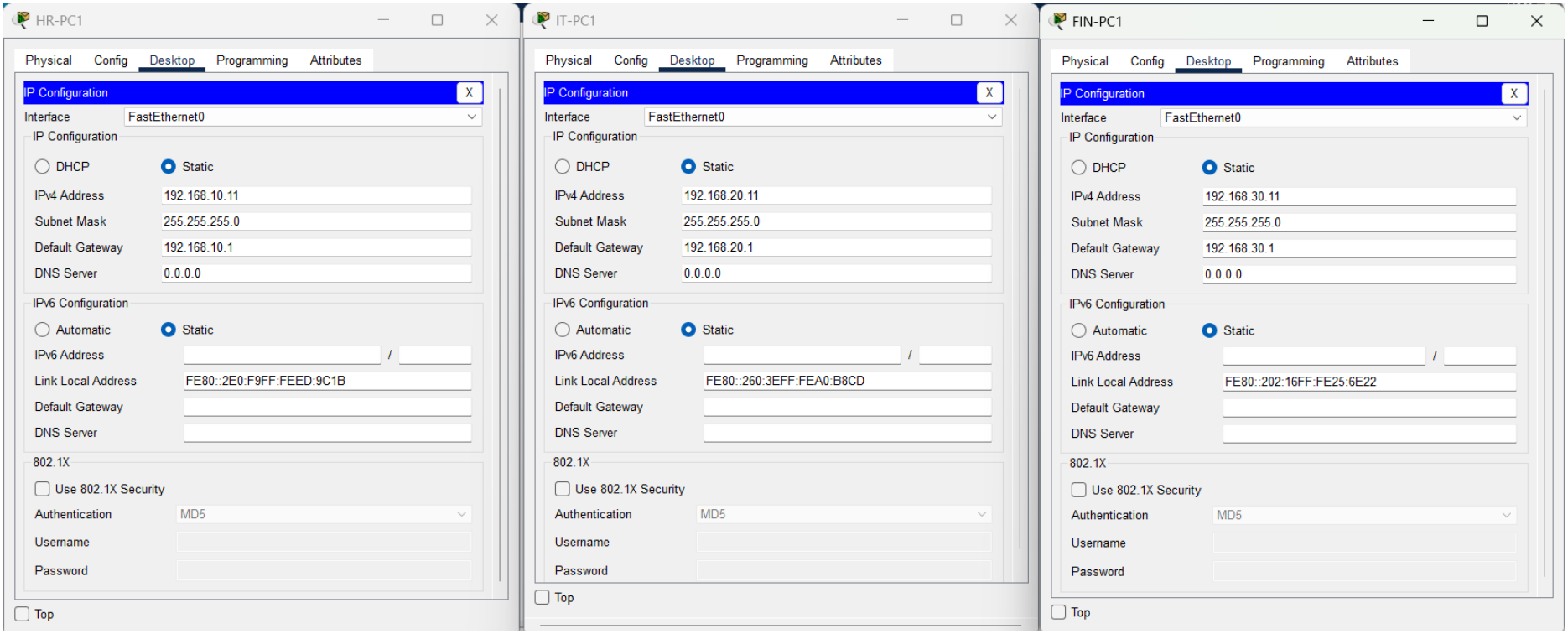


Figure B.12: IP configuration of HR-PC1, IT-PC1 and FIN-PC1

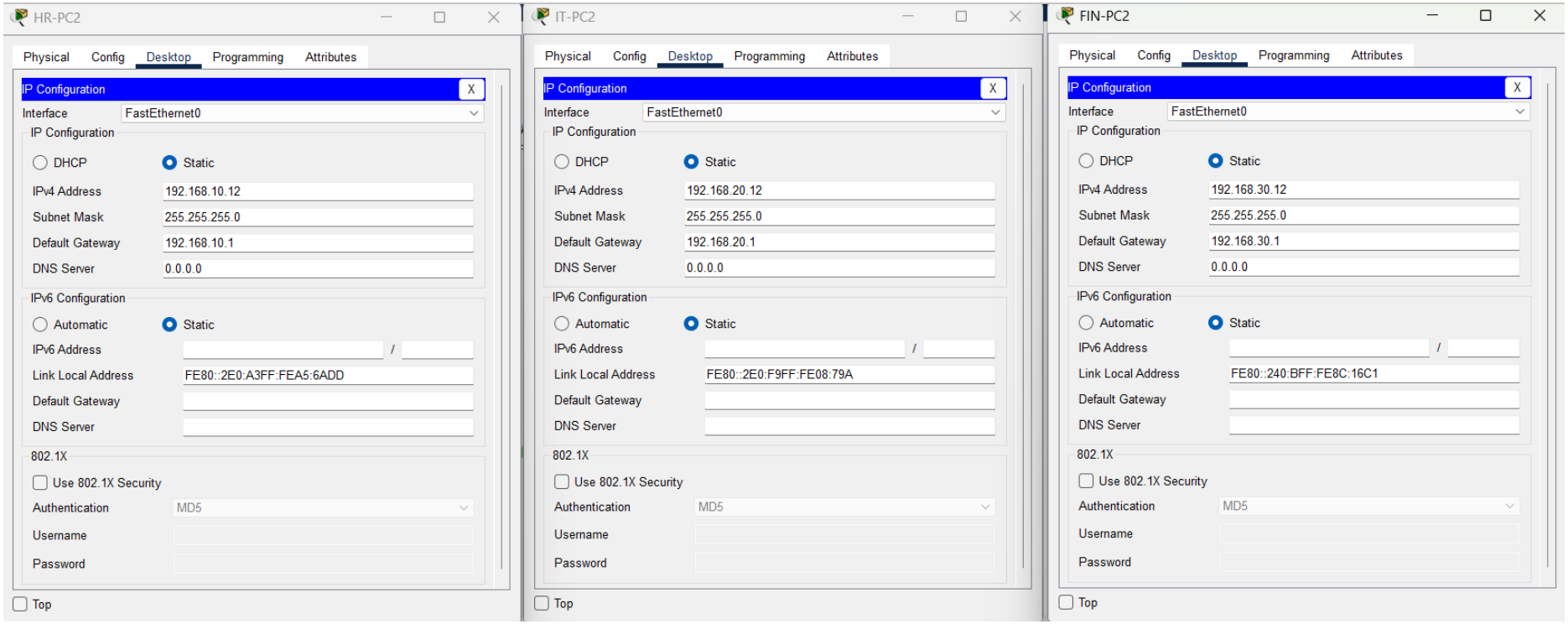


Figure B.13: IP configuration of HR-PC2, IT-PC2 and FIN-PC2

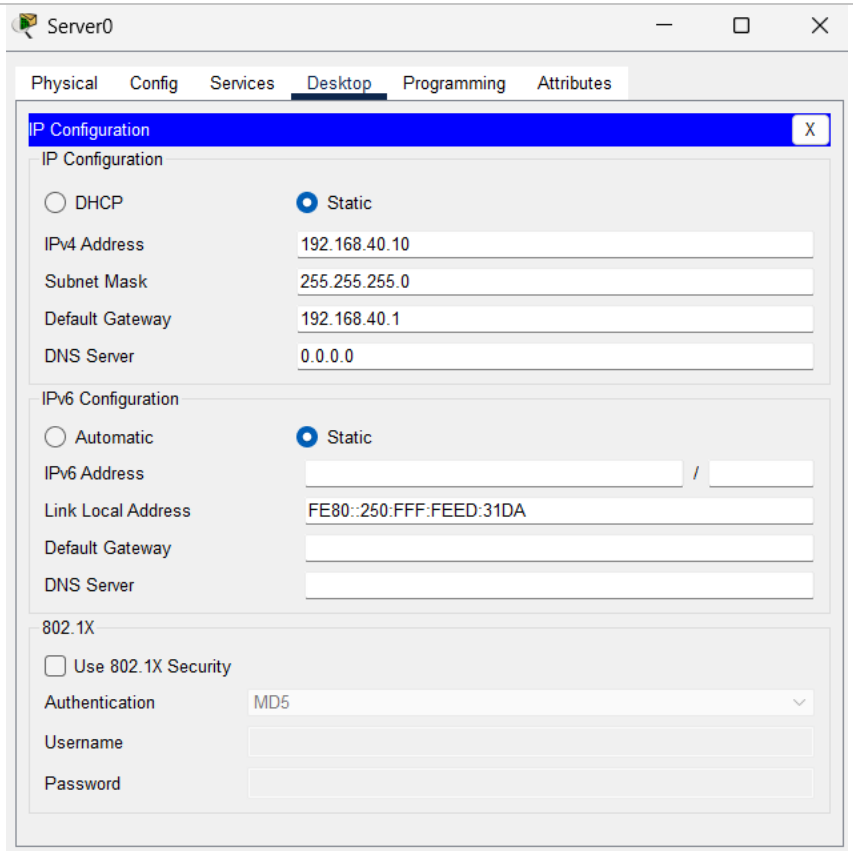


Figure B.14: IP configuration of Server0